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Storage controller and storage controller control method

Abstract

To replace a storage controller without stopping a host and without losing data related to IO processing from the host. A storage controller sets, in a port management table, a first host path definition between the host and first address information in a controller unit in addition to a second host path definition between the host and second address information in a controller unit. The storage controller sets, in a route management table, a first connection route between an input port and a first output port to which a port of the first address information is connected, in addition to a second connection route between an input port and a second output port to which a port of the second address information is connected. The storage controller transfers an IO to one controller unit or another controller unit based on the port management table and the route management table.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a storage controller and a storage controller control method.

2. Description of Related Art

(2) In the related art, in a storage device, storage controllers for processing input and output (IO) from a host are redundantly configured. By redundantly configuring a connection between the host and the storage controllers, when an abnormality occurs in one of the storage controllers, IO processing can be continued by another storage controller.

(3) For example, PTL 1 discloses a technique in which, when one of storage control devices redundantly connected to a server device fails, a connection path between the server device and the

failed storage control device is switched to a new connection path between the server device and a non-failed storage control device. By switching the connection path in this manner, it is possible to limit an influence of disconnection of the connection path when the failed storage control device is restored by restarting.

CITATION LIST

Patent Literature

(4) PTL 1: JP2017-010390A

SUMMARY OF THE INVENTION

(5) The storage controller may be removed and replaced with a new storage controller for performance improvement and failure replacement. In a storage system in which storage controllers are redundantly connected, by applying the technique disclosed in PTL 1 described above, IO to one failed storage controller can be aggregated in the other non-failed storage controller. In this manner, by replacing the other failed storage controller, the storage controller can be replaced while continuing the IO processing from the host.

(6) However, in the technique disclosed in PTL 1, the storage controller can be replaced while continuing the IO processing from the host, but disconnection may occur when the connection path between the host and the storage controller to be replaced is switched, which causes data related to the IO processing to be lost.

(7) When a connection between a host of a client and the storage controller cannot be redundantly configured, the storage controller cannot be replaced by applying the technique disclosed in PTL 1. That is, after stopping the host, the storage controller is replaced.

(8) The invention is made in view of the above, and an object of the invention is to enable replacement of the storage controller without stopping the host and without losing the data related to the IO from the host.

(9) As one aspect solving the above problems, a storage controller includes a switch unit configured to receive an IO from a host, and a first storage control device and a second storage control device configured to process the IO received by the switch unit and to output and receive data to and from a disc device. The switch unit includes an input port configured to receive the IO from the host, and an output port configured to output the IO to a port of the first storage control device and a port of the second storage control device. A storage unit of the storage controller stores port management information for managing host path definition information for each of the first storage control device and the second storage control device, the host path definition information indicating a correspondence relation between the host and address information held by the port of a transmission destination to which the host transmits the IO, and route management information for managing a connection route for connecting the input port to the output port connected to the port holding the address information in the switch unit to transfer the IO. A processor of the storage controller receives an instruction to stop the second storage control device; converts second address information to first address information, the second address information indicating the address information held by the port of the second storage control device, and the first address information indicating the address information held by the port of the first storage control device; sets, in the port management information, a first host path definition indicating the host path definition information between the host and the first address information in the first storage control device, in addition to a second host path definition indicating the host path definition information between the host and the second address information in the second storage control device; sets, in the route management information, a first connection route indicating the connection route between the input port and a first output port serving as the output port to which the port holding the first address information is connected, in addition to a second connection route indicating the connection route between the input port and a second output port serving as the output port to which the port holding the second address information is connected; and controls the switch unit based on the route management information and controls the first storage control device and the second storage

control device based on the port management information to transfer the IO to the first storage control device or the second storage control device.
(10) According to the invention, the storage controller can be replaced without stopping the host and without losing the data related to the IO from the host.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a diagram showing a configuration of a storage system according to Embodiment 1;
(2) FIG. **2**A is a diagram showing a configuration of a port management table according to Embodiment 1;
(3) FIG. **2**B is a diagram showing a configuration of a route management table according to Embodiment 1;
(4) FIG. **2**C is a diagram showing a configuration of an OX_ID management table according to Embodiment 1;
(5) FIG. **3**A is a diagram showing a storage controller before switching a connection route to a controller unit according to Embodiment 1;
(6) FIG. **3**B is a diagram showing a state of the port management table according to Embodiment 1;
(7) FIG. **3**C is a diagram showing a state of the route management table according to Embodiment 1;
(8) FIG. **4**A is a diagram showing the storage controller during a switching operation (first half) of the connection route to the controller unit according to Embodiment 1;
(9) FIG. **4**B is a diagram showing a state of the port management table according to Embodiment 1;
(10) FIG. **4**C is a diagram showing a state of the route management table according to Embodiment 1;
(11) FIG. **5** is a diagram showing the storage controller during the switching operation (second half) of the connection route to the controller unit according to Embodiment 1;
(12) FIG. **6**A is a diagram showing the storage controller after completion of the switching operation of the connection route to the controller unit according to Embodiment 1;
(13) FIG. **6**B is a diagram showing a state of the port management table according to Embodiment 1;
(14) FIG. **6**C is a diagram showing a state of the route management table according to Embodiment 1;
(15) FIG. **7** is a sequence diagram showing stop processing of an old controller according to Embodiment 1;
(16) FIG. **8** is a flowchart showing switching processing according to Embodiment 1;
(17) FIG. **9** is a sequence diagram showing activation processing of a new controller according to Embodiment 1;
(18) FIG. **10**A is a diagram showing the storage controller before switching a connection route to the controller unit according to Embodiment 2;
(19) FIG. **10**B is a diagram showing a state of the port management table according to Embodiment 2;
(20) FIG. **10**C is a diagram showing a state of the route management table according to Embodiment 2;
(21) FIG. **11**A is a diagram showing the storage controller during a switching operation (first half) of a connection route in the controller unit according to Embodiment 2;
(22) FIG. **11**B is a diagram showing a state of the port management table according to Embodiment 2;
(23) FIG. **11**C is a diagram showing a state of the route management table according to

Embodiment 2;
(24) FIG. **12** is a diagram showing the storage controller during the switching operation (second half) of the connection route in the controller unit according to Embodiment 2;
(25) FIG. **13**A is a diagram showing a storage controller **1** after the switching operation of the connection route in the controller unit according to Embodiment 2;
(26) FIG. **13**B is a diagram showing a state of the port management table according to Embodiment 2;
(27) FIG. **13**C is a diagram showing a state of the route management table according to Embodiment 2;
(28) FIG. **14** is a sequence diagram showing stop processing of an old storage controller according to Embodiment 2;
(29) FIG. **15** is a flowchart showing low-load determination and path selection processing according to Embodiment 2; and
(30) FIG. **16** is a sequence diagram showing activation processing of a new storage controller according to Embodiment 2.
DESCRIPTION OF EMBODIMENTS
(31) Hereinafter, embodiments of the invention will be described in detail with reference to the drawings. In the present embodiments, the same components are denoted by the same reference signs in principle, and repeated descriptions thereof are omitted. The present embodiments are merely examples for implementing the invention, and do not limit the technical scope of the invention. The number of components in the present embodiments is not limited unless otherwise noted.
(32) In the following description, reference numerals including indexes are used when the same components are described separately, and a reference numeral excluding the indexes is used when the same components are not described separately. For example, when controller units **15***a* and **15***b* are described separately, reference numerals **15***a* and **15***b* including indexes a and b are used, and when controller units **15***a* and **15***b* are not described separately, a reference numeral **15** excluding the indexes is used.
(33) In the following description, processing performed by a program may be described. A computer performs processing determined by a program using a memory of a main storage device or the like by means of a processor (for example, a central processing unit (CPU) or a graphics processing unit (GPU)). Therefore, a subject of the processing performed by executing the program may be the processor. The processor executes the program to implement a functional unit that performs processing.
(34) Similarly, the subject of the processing performed by executing the program may be a controller, a device, a system, a computing machine, or a node including a processor therein. The subject of the processing performed by executing the program may be a calculation unit and may include a dedicated circuit that performs specific processing. The dedicated circuit is, for example, a field programmable gate array (FPGA) or an application specific integrated circuit (ASIC).
(35) In the following description, the program may be installed in the computing machine from a program source. The program source may be, for example, a program distribution server or a non-transitory storage medium readable by the computing machine. When the program source is the program distribution server, the program distribution server may include a processor and a storage resource (storage) for storing a program to be distributed, and the processor of the program distribution server may distribute the program to be distributed to another computing machine. In the embodiments, two or more programs may be implemented as one program, or one program may be implemented as two or more programs.
(36) In the following description, various types of data will be described in a table format. However, a data format is not limited to the table format, and may be another data format such as a queue, a list, and a comma separated value (CSV). Since the various types of data do not depend on

the data format, the " . . . table" can be expressed as " . . . information".

Embodiment 1

(37) In Embodiment 1, IO to a controller to be upgraded (an old generation controller is removed and a new generation controller is inserted) of two controllers is offloaded to a port of the other controller having a lowest load. Then, the controller to be upgraded is stopped, the controller to be upgraded is removed, and an upgraded controller is activated.

(38) In the present embodiment, "IO" refers to transmission of a request and a command for input and output of data from a host to a storage device, and transmission and reception of data between the host and the storage device in response to the request or the command.

(39) Configuration of Storage System S

(40) FIG. **1** is a diagram showing a configuration of a storage system S according to Embodiment 1. The storage system S includes a storage device **100** and hosts **3**. The storage device **100** and the hosts **3** are connected via a network N. The network N is a storage area network (SAN) including one or more switches or a communication network including only a connection cable without a network device.

(41) The storage device **100** includes a storage controller **1** and disc devices **2***a* and **2***b*. FIG. **1** shows a hierarchical structure in which the disc device **2***a* is connected to the storage device **100** and the disc device **2***b* is connected to the disc device **2***a*. However, the present invention is not limited thereto, and the disc devices **2** in the storage device **100** may have another connection configuration.

(42) The storage controller **1** includes a switch unit **11** and two or more controller units **15***a* and **15***b*. The controller unit **15***a* is an example of a first storage control device. The controller unit **15***b* is an example of a second storage control device.

(43) The switch unit **11** includes a CPU **12**, a cache memory **13**, and a plurality of ports **14**. The CPU **12** executes a switching program **121**. The cache memory **13** is a volatile or nonvolatile storage unit and stores a route management table **131**.

(44) The plurality of ports **14** are communication ports for connecting the network N and the controller units **15**. The ports **14** include input ports **14**A (ports #**1** to #**8**) connected to the network N and output ports **14**B (ports #**9** to #**16**) connected to the controller units **15**.

(45) The switching program **121** switches connections of the input ports **14**A (ports #**1** to #**8**) and the output ports **14**B (ports #**9** to #**16**). The switching program **121** manages connection information indicating connection routes of the input ports **14**A (ports #**1** to #**8**) and the output ports **14**B (ports #**9** to #**16**) in the route management table **131**.

(46) The controller unit **15** includes a micro processor package board (MPB) **16**, a cache memory **17**, a connection unit **18**, a plurality of ports **19**, and a disc adapter **20**. In the present embodiment, the storage controller **1** includes two controller units **15***a* and **15***b*.

(47) The MPB **16** is implemented by a board on which a processor such as a CPU and a memory are mounted, and controls the ports **19**, the disc adapter **20** and the cache memory **17**. The CPU of the MPB **16** executes a control program **161**.

(48) The cache memory **17** is a volatile or nonvolatile storage unit, and stores data received from the hosts **3** and data read from the disc devices **2**. The cache memory **17** also stores a port management table **171**. The controller units **15***a* and **15***b* share the port management table **171**.

(49) The cache memory **13** of the switch unit **11** and the cache memory **17** of the controller unit **15** are examples of the storage unit in the storage controller **1**, and the route management table **131** and the port management table **171** may be arranged in any cache memory.

(50) The connection unit **18** connects the MPB **16**, the cache memory **17**, the ports **19**, and the disc adapter **20** to each other. The connection unit **18** communicates with a connection unit **18** of the other controller unit **15** other than the controller unit **15** in which the connection unit **18** itself is arranged, and causes the controller unit **15** in which the connection unit **18** itself is arranged to access each component of the other controller unit **15**. The connection unit **18** can be configured as

a high-speed bus such as a crossbar switch that performs data transmission by a high-speed switching operation, for example.

(51) The plurality of ports **19** are communication ports for connection with the switch unit **11**. The controller unit **15** performs data communication with an external device via the ports **19**. Address information (for example, WWN) for identifying each port **19** in the network N is assigned to each port **19**. The address information on the port **19** is a world wide name (WWN) when the network N is a SAN such as a fibre channel, a serial attached SCSI (SAS), or an iSCSI. However, the address information is not limited thereto, and may be any information that can identify the port **19** in the network N.

(52) The disc adapter **20** communicates with the disc device **2** via a communication port (not shown). The disc adapter **20** is configured as a microcomputer including a CPU, a memory, and the like. The disc adapter **20** writes data written to the cache memory **17** to a disc **21** of the disc device **2** via the plurality of ports **19**, and writes the data read from the disc **21** to the cache memory **17**.

(53) The disc adapter **20** converts a logical address of the data to a physical address when the data is input or output.

(54) The disc device **2** includes a plurality of discs **21** such as hard disc drives (HHD) and solid state drives (SSD) arranged in an array, a flexible disc, and an optical disc, and an I/O interface **22**. A magnetic tape or a semi-conductor memory (for example, a flash memory) may be used as the disc **21**.

(55) The I/O interface **22** is an interface for communicating with the controller unit **15** or another disc device **2**.

(56) The host **3** is a computer device including a host bus adapter **33** and an information processing resource such as a CPU and a memory. The host **3** issues the I/O of the data to the storage device **100** via the host bus adapter by an application **31** running on an operating system (OS) **32** running on the CPU.

(57) In the storage controller **1**, the CPU **12** of the switch unit **11** and the CPU mounted in the MPB **16** of the controller unit **15** can communicate with each other, and can refer to the route management table **131** and the port management table **171** as necessary.

(58) The CPU **12** of the switch unit **11** and the CPU mounted in the MPB **16** of the controller unit **15** are examples of a processor in the storage controller **1**. Each processing function of the CPU **12** and the CPU mounted on the MPB **16** described in the present embodiment can be appropriately changed in design to be able to be performed by any CPU.

(59) Configuration of Port Management Table **171**

(60) FIG. **2**A is a diagram showing a configuration of the port management table **171** according to Embodiment 1. The port management table **171** has columns of “CTL #”, “Port”, “Address”, “load”, “host definition”, and “LU definition”.

(61) The “CTL #” is identification n information on the controller unit **15**. The “Port” is identification information on the ports **14** (ports #**9** to #**16**) on a controller unit **15** side of the switch unit **11** connected to the port **19** of a corresponding record. The “Address” indicates a WWN of the port **19** identified by the “CTL #” and the “Port”.

(62) The “load” indicates a load of the port **19** identified by the “CTL #” and the “Port”. The “host definition” is identification information on the host **3** connected to the controller unit **15** via the port **19** identified by the “CTL #” and the “Port”. The “LU definition” is identification information in a logical storage area of the disc device **2** connected via the port **19** identified by the “CTL #” and the “Port”.

(63) That is, the port management table **171** manages, for each of the controller units **15***a* and **15***b*, host path definition information indicating a correspondence relation between the host **3** and the address information (WWN) held by the port **19**, which is a transmission destination to which the host **3** transmits the IO.

(64) Configuration of Route Management Table **131**

(65) FIG. **2**B is a diagram showing a configuration of the route management table **131** according to Embodiment 1. The route management table **131** includes columns of “Port”, “storage target WWN”, “connection route”, and “presence or absence of conversion”.
(66) The “Port” is identification information of the output ports **14**B (ports #**9** to #**16**). The “storage target WWN” indicates the WWN of the port **19** to which the output port **14**B identified by the “Port” is connected. The “storage target WWN” is blank when the connection between the output port **14**B identified by “Port” and the port **19** is not defined.
(67) The “connection route” stores connection information indicating a connection route between the input ports **14**A (ports #**1** to #**8**) and the output ports **14**B (ports #**9** to #**16**) when the connection route is defined. The “connection route” is blank when the connection route is not defined.
(68) The “presence or absence of conversion” stores “presence of conversion” when the corresponding record indicates the connection route after the conversion of the “storage target WWN” in the switch unit **11**, and the “presence or absence of conversion” is blank when the record indicates the connection route before the conversion.
(69) That is, the route management table **131** manages, in the switch unit **11**, a connection route for connecting the input port **14**A and the output port **14**B connected to the port **19** having the address information (WWN) to transfer the IO.
(70) Configuration of OX_ID Management Table **132**
(71) FIG. **2**C is a diagram showing a configuration of the OX_ID management table **132** according to Embodiment 1. The OX_ID management table **132** is stored in the cache memory **13** of the switch unit **11** and manages an uncompleted IO from the host **3**. The OX_ID management table **132** includes columns of “Port”, “originator exchange_identifier (OX_ID) of uncompleted IO”, “completion flag”, and “deletable”.
(72) The “Port” indicates identification information of the ports **14** (ports #**1** to #**8**) on a network N side whose connection route to the ports **14** (ports #**9** to #**16**) on the controller unit **15** side is switched during processing of the IO input from the host **3**.
(73) The “OX_ID of uncompleted IO” is identification information of the uncompleted IO. As for the “completion flag”, “completed” is input when the corresponding uncompleted IO is completed, and “uncompleted” is input when the corresponding uncompleted IO is uncompleted. The “deletable” indicates whether the corresponding record in the OX_ID management table **132** is deletable.
(74) Operation Outline of Storage Controller **1** according to Embodiment 1
(75) FIGS. **3**A to **6**C are diagrams showing an operation outline of the storage controller **1** and states of the port management table **171** and the route management table **131** at a time of each operation according to Embodiment 1. In Embodiment 1, the controller unit **15***b* of the controller units **15***a* and **15***b* is to be upgraded. That is, the I/O to the controller unit **15***b* is offloaded to the controller unit **15***a* to stop the controller unit **15***b*, the controller unit **15***b* is upgraded to a controller unit **15***b***1**, and then the controller unit **15***b***1** is activated.
(76) Before Switching of Connection Route to Controller Unit **15***a*
(77) FIG. **3**A is a diagram showing the storage controller **1** before switching the connection route to the controller unit **15***a*. The port management table **171** before switching the connection route is as shown in FIG. **3**B. The route management table **131** before switching the connection route is as shown in FIG. **3**C.
(78) In the state shown in FIG. **3**A, host path definitions **162***a*, **162***b*, **162***c*, and **162***d* are defined in the controller unit **15***a*. A host path definition **163** is defined in the controller unit **15***b*. Host path definitions **162***a*, **162***b*, **162***c*, **162***d*, and **163** correspond to records **1711**, **1712**, **1713**, **1714**, and **1715** in the port management table **171** (FIG. **3**B), respectively.
(79) During Switching Operation (First Half) of Connection Route to Controller Unit **15***a* according to Embodiment 1
(80) FIG. **4**A is a diagram showing the storage controller **1** during the switching operation (first

half) of the connection route to the controller unit **15***a* according to Embodiment 1. The port management table **171** during the switching operation (first half) of the connection route is as shown in FIG. **4**B. The route management table **131** during the switching operation (first half) of the connection route is as shown in FIG. **4**C.
(81) The control program **161** of the controller unit **15***b* outputs, to the controller unit **15***a*, an update instruction for setting the host path definition **163** in the controller unit **15***a*. The control program **161** of the controller unit **15***a* sets the host path definition **163** in one of the plurality of ports **19** of the controller unit **15***a* within a range not exceeding an upper limit of the load of each port **19**. In an example of FIG. **4**A, the control program **161** of the controller unit **15***a* sets the host path definition **163** in the port **19** (WWN #**4**) having a lowest port load among the plurality of ports **19** of the controller unit **15***a*.
(82) The control program **161** of the controller unit **15***a* reflects, in the port management table **171**, the host path definition **163** additionally set in the controller unit **15***a*. The host path definition related to the port **19** (WWN #**4**) to which the host path definition **163** is additionally set corresponds to a record **1716** in the port management table **171** (FIG. **4**B).
(83) The "load" of the records **1716** and **1717** is "100 MB/s" because the host path definition **163** related to the record **1715** (FIGS. **3**B and **4**B) is additionally set to the "WWN #**4**". "100 MB/s" is the sum of "load" of "20 MB/s" in "WWN #**3**" and "load" of "80 MB/s" in "WWN #**6**". In the port management table **171** shown in FIG. **4**B, the record **1715** is maintained as in FIG. **3**B.
(84) The control program **161** of the controller unit **15***b* outputs an instruction to switch the connection route to the switching program **121** of the switch unit **11**. According to the switching instruction, the switching program **121** additionally sets a connection route (the input port **14**A (port #**6**) to the output port **14**B (port #**12**)) according to the setting of the host path definition **163** to the output port **14**B (ports #**9** to #**12**) on the controller unit **15***a* side. The switch unit **11** transfers the host IO input via the input port **14**A (port #**6**) to the controller unit **15***a* by the additionally set connection route and WWN conversion (WWN #**6** to WWN #**4**).
(85) After completing the switching of the connection route, the switching program **121** reflects the switching of the connection route in the route management table **131**. The connection route related to the port **19** (WWN #**4**) to which the host path definition **163** is additionally set corresponds to a record **1312** of the route management table **131** (FIG. **4**C).
(86) The record **1312** indicates that the connection route related to the record **1311** (FIGS. **3**C and **4**C) is added as the connection route of the input port **14**A (port #**6**) to the output port **14**B (port #**12**) by the WWN conversion (WWN #**6** to WWN #**4**). Since the record **1312** involves the WWN conversion, the "presence or absence of conversion" is the "presence of conversion". In the route management table **131** shown in FIG. **4**C, the record **1311** is maintained as in FIG. **3**C.
(87) During Switching Operation (Second Half) of Connection Route to Controller Unit **15***a* according to Embodiment 1
(88) FIG. **5** is a diagram showing the storage controller **1** during the switching operation (second half) of the connection route to the controller unit **15***a* according to Embodiment 1.
(89) FIG. **5** shows that an IO (new IO) newly input from the input port **14**A (port #**6**) after the start of the switching operation of the connection route to the controller unit **15***a* shown in FIG. **4**A is allocated to the controller unit **15***a* for processing. The connection route when the IO is allocated to the controller unit **15***a* is from the input port **14**A (port #**6**) to the output port **14**B (port #**12**).
(90) FIG. **5** shows that an I/O (IO in process), in which the processing input from the input port **14**A (port #**6**) before the start of the switching operation of the connection route to the controller unit **15***a* shown in FIG. **4**A is uncompleted, is allocated to the controller unit **15***b* for processing. The connection route when the IO is allocated to the controller unit **15***b* is from the input port **14**A (port #**6**) to the output port **14**B (port #**14**).
(91) After Completing Switching Operation of Connection Route to Controller Unit **15***a* according to Embodiment 1

(92) FIG. **6**A is a diagram showing the storage controller **1** after completing the switching operation of the connection route to the controller unit **15***a* according to Embodiment 1. FIG. **5** shows the storage controller **1** in a state in which all the I/Os in process are allocated to the controller unit **15***b* and processing is completed.
(93) The port management table **171** after the switching operation of the connection route is completed is as shown in FIG. **6**B. The route management table **131** after the switching operation of the connection route is completed is as shown in FIG. **6**C.
(94) In the state shown in FIG. **6**A, the control program **161** of the controller unit **15***b* executes definition deletion of deleting the host path definition **163** in the controller unit **15***b*, and reflects the deletion in the port management table **171**. The host path definition related to the port **19** (WWN #**6**) from which the host path definition **163** is deleted corresponds to the record **1718** in the port management table **171** (FIG. **6**B). As shown in the record **1718**, the host path definition **163** related to the record **1715** (FIGS. **3**B and **4**B) is clear such that the "load" is "0", and the "host definition" and the "LU definition" are "undefined". Thereafter, the control program **161** of the controller unit **15***b* closes (shuts down) the controller unit **15***b*.
(95) The control program **161** of the controller unit **15***b* outputs an instruction to release the connection route to the switching program **121** of the switch unit **11**. The switching program **121** deletes the connection route (the input port **14**A (port #**6**) to the output port **14**B (port #**14**)) according to the deletion of the host path definition **163** in accordance with the release instruction. Thereafter, the switch unit **11** transfers the host IO input via the input port **14**A (port #**6**) to only the controller unit **15***a* through the connection route (the input port **14**A (port #**6**) to the output port **14**B (port #**12**)).
(96) The control program **161** reflects the release of the connection route in the route management table **131**. The released connection route corresponds to the record **1313** in the route management table **131** (FIG. **6**C). As shown in the record **1313**, the connection route related to the record **1311** (FIGS. **3**C and **4**C) is clear such that the "connection route" and the "presence or absence of conversion" are blank.
(97) Stop Processing Sequence of Old Controller according to Embodiment 1
(98) FIG. **7** is a sequence diagram showing stop processing of an old controller according to Embodiment 1.
(99) First, in step S**101**, a terminal t**1** capable of communicating with the storage controller **1** via the network N receives an instruction to stop the old controller from an operator. In the present embodiment, the old controller to be upgraded is the controller unit **15***b*. Next, in step S**102**, the terminal t**1** transmits the received instruction to stop the old controller to the control program **161** of the controller unit **15***a* on the counterpart side of the old controller.
(100) Next, in step S**103**, the control program **161** of the controller unit **15***a* on the counterpart side starts stopping the old controller. Next, in step S**104**, the control program **161** of the controller unit **15***a* transmits the instruction to stop the old controller to the control program **161** of the controller unit **15***b*.
(101) Next, in step S**105**, the control program **161** of the controller unit **15***b* starts switching the host path definition **163**. Next, in step S**106**, the control program **161** of the controller unit **15***b* transmits an instruction to update the host path definition **163** to the control program **161** of the controller unit **15***a*.
(102) Next, in step S**107**, the control program **161** of the controller unit **15***a* sets, in the controller unit **15***a*, the host path definition **163** in the port management table **171** in response to the update instruction of the host path definition **163**. Specifically, as shown in FIG. **4**B, the port **19** (WWN #**4**) having the lowest load of the controller unit **15***a* is selected. The record **1716** in which the host path definition **163** of the record **1715** (FIG. **3**B) is set in the record **1714** (FIG. **3**B) is generated such that the host path definition **163** is set in the selected port **19**, and the "load" of the record **1717** is updated.

(103) Next, in step S**108**, the control program **161** of the controller unit **15***a* notifies the control program **161** of the controller unit **15***b* of completion of table update in step S**107**.
(104) Next, in step S**109**, the control program **161** of the controller unit **15***b* executes connection route switching in the switch unit **11**. Next, in step S**110**, the control program **161** of the controller unit **15***b* transmits the instruction to switch the connection route in the switch unit **11** to the switch unit **11**.
(105) Next, in step S**111**, the switching program **121** of the switch unit **11** sets a new connection route in response to the instruction to switch the connection route. Next, in step S**112**, the switching program **121** of the switch unit **11** reflects information of the new connection route in the route management table **131**. Specifically, as shown in FIG. **4**C, the record **1312** is generated in the route management table **131**.
(106) Next, in step S**113**, the switching program **121** of the switch unit **11** executes switching processing. Details of the switching processing will be described later with reference to FIG. **8**.
(107) Next, in step S**114**, the switching program **121** of the switch unit **11** deletes an existing connection route that overlaps between the connection route newly set in step S**111** and the host **3** of an IO source. Specifically, in the route management table **131**, the "connection route" and the "presence or absence of conversion" of the record **1311** having the same WWN before conversion of the record **1312** with the "presence of conversion" for the "presence or absence of conversion" shown in FIG. **4**C are cleared as in the record **1313** shown in FIG. **6**C.
(108) Next, in step S**115**, the switching program **121** of the switch unit **11** transmits a notification of switching completion of the connection route to the control program **161** of the controller unit **15***b*.
(109) Next, in step S**116**, the control program **161** of the controller unit **15***b* executes closing processing of the controller unit **15***b*. Next, in step S**117**, the control program **161** of the controller unit **15***b* notifies the control program **161** of the controller unit **15***a* of closing completion of the controller unit **15***b*.
(110) Next, in step S**118**, the control program **161** of the controller unit **15***a* receives the notification of closing completion of the controller unit **15***b* from the control program **161** of the controller unit **15***b*. Next, in step S**119**, the control program **161** of the controller unit **15***a* notifies the terminal t**1** of the closing completion of the controller unit **15***b*.
(111) Next, in step S**120**, the terminal t**1** outputs the closing completion of the controller unit **15***b* to a display screen or the like, and ends the stop processing of the old controller.
(112) Switching Processing According to Embodiment 1
(113) FIG. **8** is a flowchart showing the switching processing (step S**113** (FIG. **7**)) according to Embodiment 1.
(114) First, in step S**113***a*, the switching program **121** of the switch unit **11** adds the OX_ID of the uncompleted IO currently being processed to the OX_ID management table **132** (FIG. **2**C). Next, in step S**113***b*, the switching program **121** determines whether the input IO is a new IO. The switching program **121** proceeds the processing to step S**113***g* if the input IO is a new IO (YES in step S**113***b*), and proceeds the processing to step S**113***c* if the input IO is an uncompleted IO (NO in step S**113***b*).
(115) In step S**113***c*, the switching program **121** processes the uncompleted IO in a current controller before the switching. The current controller is the controller unit **15***b* when the switching processing is executed in the stop processing of the old controller in FIG. **7**.
(116) Next, in a step S**113***d*, when receiving the IO completion notification of the uncompleted IO from the current controller, the switching program **121** deletes the OX_ID of the uncompleted IO from the OX_ID management table **132**.
(117) Next, in step S**113***e*, the switching program **121** checks whether there is the OX_ID of the uncompleted IO in the OX_ID management table **132**. Next, in step S**113***f*, when there is the OX_ID of the uncompleted IO in the OX_ID management table **132** (YES in step S**113***f*), the

switching program **121** returns the processing to step S**113***b*. On the other hand, when there is no OX_ID of the uncompleted IO (NO in step S**113***f*), the switching processing ends, and the processing proceeds to step S**114** (FIG. **7**).
(118) On the other hand, in step S**113***g*, the switching program **121** processes the uncompleted IO in a switching destination controller. The switching destination controller is the controller unit **15***a* when the switching processing is executed in the stop processing of the old controller in FIG. **7**.
(119) Next, in step S**113***h*, the switching program **121** receives the IO completion notification. When the step S**113***h* ends, the switching program **121** proceeds the processing to step S**113***f*.
(120) Activation Processing Sequence of New Controller according to Embodiment 1
(121) FIG. **9** is a sequence diagram showing activation processing of the new controller according to Embodiment 1. The activation processing of the new controller is executed after the old controller is upgraded to the controller unit **15***b***1** after ending the stop processing of the old controller (FIG. **7**).
(122) First, in step S**121**, the terminal t**1** receives an instruction to activate the new controller from the operator. In the present embodiment, the new controller is the controller unit **15***b***1**. Next, in step S**122**, the terminal t**1** transmits the received instruction to activate the new controller to the control program **161** of the controller unit **15***a* on the counterpart side of the new controller.
(123) Next, in step S**123**, the control program **161** of the controller unit **15***a* on the counterpart side starts the activation of the new controller. Next, in step S**124**, the control program **161** of the controller unit **15***a* transmits the instruction to activate the new controller to the control program **161** of the controller unit **15***b***1**.
(124) Next, in step S**125**, the control program **161** of the controller unit **15***b***1** activates the controller unit **15***b***1** serving as the new controller. In step S**126**, the control program **161** of the controller unit **15***b***1** starts switching the host path definition **163**. Here, as shown in FIG. **6**A, the host path definition **163** to be switched is the host path definition **163** that is moved from the controller unit **15***b* to the controller unit **15***a*.
(125) Next, in step S**127**, the control program **161** of the controller unit **15***b***1** acquires the host path definition **163** from the port management table **171**. Next, in step S**128**, the control program **161** of the controller unit **15***b***1** transmits an instruction to update the host path definition **163** to the control program **161** of the controller unit **15***a*.
(126) Next, in step S**129**, the control program **161** of the controller unit **15***a* sets the host path definition **163** to the controller unit **15***b***1** in the port management table **171** in response to the instruction to update the host path definition **163**. That is, the port management table **171** is updated such that the host path definition **163** moved from the controller unit **15***b* to the controller unit **15***a* in step S**107** in the stop processing of the old controller in FIG. **7** is returned from the controller unit **15***a* to the controller unit **15***b***1**. Specifically, the port management table **171** is updated from the state in FIG. **6**B to the state in which the record **1715** in FIG. **4**B is restored.
(127) Next, in step S**130**, the control program **161** of the controller unit **15***a* notifies the control program **161** of the controller unit **15***b***1** of the completion of the table update in step S**129**.
(128) Next, in step S**131**, the control program **161** of the controller unit **15***b***1** executes the connection route switching in the switch unit **11**. Next, in step S**132**, the control program **161** of the controller unit **15***b***1** acquires the corresponding record with the “presence of conversion” for the “presence or absence of conversion” in the route management table **131** from the switch unit **11** as route information.
(129) Next, in step S**133**, the control program **161** of the controller unit **15***b***1** transmits, to the switch unit **11**, the instruction to switch the connection route in the switch unit **11** based on the route information acquired in step S**132**.
(130) Next, in step S**134**, the switching program **121** of the switch unit **11** sets the new connection route in response to the instruction to switch the connection route. The new connection route set here is, for example, the connection route indicated by the record **1311** in the route management

table **131** shown in FIG. **4**C obtained by restoring the record **1313** in the route management table **131** shown in FIG. **6**C to the state before the switching processing.
(131) Next, in step S**135**, the switching program **121** of the switch unit **11** reflects the information of the new connection route set in step S**134** in the route management table **131**. Specifically, the record **1311** is generated in the route management table **131** shown in FIG. **4**C.
(132) Next, in step S**136**, the switching program **121** of the switch unit **11** executes the switching processing. Details of the switching processing in step S**136** are the same as those in step S**113**. In the switching processing in step S**136**, the current controller is the controller unit **15***a*, and the switching destination controller is the controller unit **15***b***1**.
(133) Next, in step S**137**, the switching program **121** of the switch unit **11** deletes an existing connection route that overlaps between the connection route newly set in step S**134** and the host **3** of the IO source in the route management table **131**. Specifically, the record **1312** is deleted in the route management table **131** shown in FIG. **4**C.
(134) Next, in step S**138**, the switching program **121** of the switch unit **11** transmits the notification of switching completion of the connection route to the control program **161** of the controller unit **15***b*.
(135) Next, in step S**139**, the control program **161** of the controller unit **15***b***1** receives the notification of switching completion from the switching program **121** of the switch unit **11**, and completes the activation of the controller unit **15***b*. In step S**140**, the control program **161** of the controller unit **15***b***1** notifies the control program **161** of the controller unit **15***a* of activation completion of the controller unit **15***b***1**.
(136) Next, in step S**141**, the control program **161** of the controller unit **15***a* receives the activation completion of the controller unit **15***b***1** from the control program **161** of the controller unit **15***b***1**. Next, in step S**142**, the control program **161** of the controller unit **15***a* notifies the terminal t**1** of the activation completion of the controller unit **15***b***1**.
(137) Next, in step S**143**, the terminal t**1** outputs the activation completion of the controller unit **15***b***1** to the display screen or the like, and ends the activation processing of the new controller.
Effects of Embodiment 1
(138) In Embodiment 1 described above, a second host path definition between second address information in the controller unit **15***b* to be updated and the host **3** is set in the port management table **171**. Further, a first host path definition between first address information in the controller unit **15***a* and the host **3** is set in the port management table **171**. A second connection route between the input port **14**A and a second output port serving as the output port **14**B to which the port **19** having the second address information is connected, is set in the route management table **131**. Further, a first connection route between the input port **14**A and a first output port serving as the output port **14**B to which the port **19** having the first address information is connected, is set in the route management table **131**. In this way, the WWN address is converted, and the connection route and the host path definition information are double set such that the IO to one controller unit **15** is transferred to both controller units **15**. According to the port management table **171** and the route management table **131**, the IO is transferred to the controller unit **15***a* or **15***b*.
(139) Thus, according to Embodiment 1, the storage controller **1** of the storage device **100** includes the controller unit **15** including a redundant configuration and an interface capable of switching the connection route. Storage target information (WWN) visible from the host **3** is maintained by switching the connection route by the interface. As a result, the controller unit **15** can be upgraded without stopping the host **3** and deleting the data related to the IO from the host **3**.
(140) In Embodiment 1 described above, identification information of the IO in process is recorded in the OX_ID management table **132**, and it is determined whether the identification information of the IO is recorded in the OX_ID management table **132** for each IO. When the identification information of the IO is recorded in the OX_ID management table **132**, the IO in process is output to the controller unit **15***b* via the second connection route. When the identification information of

the IO is not recorded in the OX_ID management table **132**, the new IO is output to the controller unit **15***a* via the first connection route.
(141) Thus, according to Embodiment 1, it is possible to avoid the occurrence of instantaneous interruption of the IO by distinguishing between the new IO and the IO in process and switching the controller unit **15** of the transfer destination.
Embodiment 2
(142) In Embodiment 2, before the IO to the controller to be upgraded is offloaded to another controller, the IO is aggregated in advance in the other controller to create a free resource. In this way, the IO to the controller to be upgraded is moved to the free resource ensured, thereby preventing a migration load from becoming excessive.
(143) In the description of Embodiment 2, repeated description with Embodiment 1 will be omitted, and differences from Embodiment 1 will be mainly described.
(144) Operation Outline of Storage Controller **1**B according to Embodiment 2
(145) FIGS. **10**A to **13**C are diagrams showing an operation outline of the storage controller **1** according to Embodiment 2 and states of the port management table **171** and the route management table **131** at a time of each operation. In Embodiment 2, IO connection paths are aggregated in the controller unit **15***a* on the counterpart side of the controller unit **15***b* to be upgraded to create the free resource of the port **19**. The IO to the controller unit **15***b* is offloaded to the free resource of the controller unit **15***a* to stop the controller unit **15***b*, the controller unit **15***b* is upgraded to a controller unit **15***b***1**, and then the controller unit **15***b***1** is activated.
(146) FIG. **10**A is a diagram showing the storage controller **1** before switching the connection route of the controller unit **15**. The port management table **171** before switching the connection route is as shown in FIG. **10**B. The route management table **131** before switching the connection route is as shown in FIG. **10**C.
(147) In a state shown in FIG. **10**A, the control program **161** of the controller unit **15***b* outputs, to the switching program **121** of the switch unit **11**, a low-load port search instruction to search for a low-load port of the counterpart controller unit **15***a*. The control program **161** of the controller unit **15***a* specifies a plurality of (for example, two) ports having the lowest load among the plurality of ports **19** of the controller unit **15***a*. In an example of FIG. **10**A, it is assumed that the control program **161** of the controller unit **15***a* specifies the port **19** (WWN #**4**) of the controller unit **15***a* as a low-load port having the lowest load. Further, it is assumed that the port **19** (WWN #**3**) of the controller unit **15***a* is specified as a low-load port having a second lowest load.
(148) During Switching Operation (First Half) of Connection Route in Controller Unit **15***a* according to Embodiment 2
(149) FIG. **11**A shows the switching operation (first half) of the connection route in the controller unit **15***a* according to Embodiment 2. The port management table **171** during the aggregation of the connection routes of the controller unit **15***a* is as shown in FIG. **11**B. The route management table **131** during the aggregation of the connection routes of the controller unit **15***a* is as shown in FIG. **11**C.
(150) As shown in FIG. **11**A, the control program **161** of the controller unit **15***a* additionally sets, to the port **19** (WWN #**3**) having the second lowest load of the controller unit **15***a*, the host path definition **162***d* related to the port **19** (WWN #**4**) having the lowest load.
(151) The control program **161** of the controller unit **15***a* reflects, in the port management table **171**, the host path definition **162***d* additionally set in the controller unit **15***a*. The host path definition related to the port **19** (WWN #**3**) to which the host path definition **162***d* is additionally set corresponds to the record **1722** in the port management table **171** (FIG. **11**B).
(152) The “load” of the records **1719** and **1722** is “35 MB/s” because the host path definition **162***d* related to the record **1720** (FIGS. **10**B and **11**B) is additionally set to “WWN #**3**”. “35 MB/s” is the sum of the “load” of the “WWN #**3**” of “20 MB/s” and the “load” of the “WWN #**4**” of “15 MB/s”. In the port management table **171** shown in FIG. **10**B, the record **1719** is maintained as in FIG.

10B.
(153) The control program **161** of the controller unit **15***b* outputs an instruction to switch the connection route to the switching program **121** of the switch unit **11**. In accordance with the switching instruction, the switching program **121** additionally sets a connection route (the input port **14**A (port #**4**) to the output port **14**B (port #**11**)) according to the setting of the host path definition **162***d* to the output port **14**B (port #**11**) on the controller unit **15***a* side. The switch unit **11** outputs the IO input via the input port **14**A (port #**4**) to the port **19** (WWN #**3**) of the controller unit **15***a* via the output port **14**B (port #**11**) by the additionally set connection route and WWN conversion (WWN #**4** to WWN #**3**).
(154) After completing the switching of the connection route, the switching program **121** reflects the switching of the connection route in the route management table **131**. The connection route related to the port **19** (WWN #**3**) to which the host path definition **162***d* is additionally set corresponds to the record **1315** in the route management table **131** (FIG. **11**C).
(155) The record **1315** indicates that the connection route related to the record **1314** (FIG. **10**C) is added as a connection route of the input port **14**A (port #**4**) to the output port **14**B (port #**11**) by the WWN conversion (WWN #**4** to WWN #**3**) in the switch unit **11**. Since the record **1315** involves the WWN conversion, the "presence or absence of conversion" is the "presence of conversion". In the route management table **131** shown in FIG. **11**C, the record **1314** is maintained as in FIG. **10**C.
(156) During Switching Operation (Second Half) of Connection Route in Controller Unit **15***a* according to Embodiment 2
(157) FIG. **12** is a diagram showing the storage controller **1** during the switching operation (second half) of the connection route in the controller unit **15***a* according to Embodiment 2.
(158) FIG. **12** shows that a command and an IO newly input from the input port **14**A (port #**4**) after the start of the aggregation processing of the connection routes in the controller unit **15***a* shown in FIG. **11**A are input to the controller unit **15***a* and processed. The connection route when being input to the controller unit **15***a* is from the input port **14**A (port #**4**) to the output port **14**B (port #**11**).
(159) FIG. **12** shows that a command and an IO that are in process and that are input from the input port **14**A (port #**4**) before the start of the aggregation processing of the connection route in the controller unit **15***a* shown in FIG. **11**A are input to the controller unit **15***a* and processed. The connection route when being input to the controller unit **15***a* is the input port **14**A (port #**4**) to the output port **14**B (port #**12**).
(160) Storage Controller **1** after Switching Operation of Connection Route according to Embodiment 2
(161) FIG. **13**A is a diagram showing the storage controller **1** after the switching operation of the connection route according to Embodiment 2. FIG. **13**A shows the storage controller **1** in a state in which all the IOs in process input via the input port **14**A (port #**4**) are allocated to the controller unit **15***b* and the processing is completed.
(162) The port management table **171** after the switching operation of the connection route of the controller unit **15***a* is as shown in FIG. **13**B.
(163) In the state shown in FIG. **13**A, the control program **161** of the controller unit **15***b* executes definition deletion of deleting the host path definition **162***d* connected to the port **19** (WWN #**4**) in the controller unit **15***a*, and reflects the deletion in the port management table **171**. The host path definition related to the port **19** (WWN #**4**) from which the host path definition **162***d* connected to the port **19** (WWN #**4**) is deleted corresponds to the record **1723** in the port management table **171** (FIG. **13**B). As shown in the record **1723**, the host path definition **162***d* related to the record **1720** (FIGS. **10**B and **11**B) is clear such that the "load" is "0", and the "host definition" and the "LU definition" are "undefined".
(164) The route management table **131** after the aggregation of the connection routes of the controller unit **15***a* is as shown in FIG. **13**C.
(165) The control program **161** reflects the release of the connection routes from the input port **14**A

(port #**4**) to the output port **14**B (port #**12**) in the route management table **131**. The released connection route corresponds to the record **1316** in the route management table **131** (FIG. **13**C). As shown in the record **1316**, the connection route related to the record **1316** (FIG. **13**C) is clear such that the "connection route" and the "presence or absence of conversion" are blank.
(166) After the control program **161** of the controller unit **15***b* sets the host path definition **163** to the controller unit **15***a* (FIG. **4**A) for the controller unit **15***a*, Embodiment 2 is the same as Embodiment 1. That is, the IO to the controller to be upgraded is offloaded to the other controller, the old controller to be upgraded is upgraded, and then the new controller is restarted.
(167) Stop Processing of Old Storage Controller according to Embodiment 2
(168) FIG. **14** is a sequence diagram showing stop processing of an old storage controller according to Embodiment 2.
(169) As compared with the stop processing of the old storage controller according to Embodiment 1 shown in FIG. **7**, the stop processing of the old storage controller according to Embodiment 2 is different in that steps S**102***a* to S**102***g* are executed between step S**102** and step S**103**. Except for the difference, the stop processing of the old storage controller according to Embodiment 2 is the same as the stop processing of the old storage controller according to Embodiment 1.
(170) In step S**102***a*, the control program **161** of the controller unit **15***a* starts switching processing of the connection route. Next, in step S**102***b*, the control program **161** of the controller unit **15***a* transmits the instruction to switch the connection route to the switching program **121** of the switch unit **11**.
(171) Next, in step S**102***c*, the switching program **121** of the switch unit **11** executes low-load determination and connection route selection processing. Details of the low-load determination and connection route selection processing will be described later with reference to FIG. **15**.
(172) Next, in step S**102***d*, the switching program **121** of the switch unit **11** reflects the connection route selected in step S**102***c* in the route management table **131** to update the route management table **131**. Next, in step S**102***e*, the switching program **121** of the switch unit **11** transmits a notification of update completion of the connection route to the control program **161** of the controller unit **15***a*. The notification of update completion of the connection route includes port information selected in step S**102***c*.
(173) Next, in step S**102***f*, the control program **161** of the controller unit **15***a* switches the host path definition based on the notification of update completion of the connection route notified in step S**102***e*, and updates the port management table **171**.
(174) Low-Load Determination and Connection Route Selection Processing according to Embodiment 2
(175) FIG. **15** is a flowchart showing the low-load determination and connection route selection processing according to Embodiment 2 shown in step S**102***c* in FIG. **14**.
(176) First, in step S**102***c***1**, the switching program **121** of the switch unit **11** acquires a performance value of a load of the output port **14**B (ports #**9** to #**12**) (controller port) connected to the controller unit **15***a* on the counterpart side. The switching program **121** acquires, for example, as the performance value of the load, a communication amount per unit time of each controller port, which is constantly acquired and accumulated by the switch unit **11**.
(177) Next, in step S**102***c***2**, the switching program **121** of the switch unit **11** updates the column "load" in the port management table **171** shared by the controller units **15***a* and **15***b* with the performance value of the load of the controller port acquired in step S**102***c***1**.
(178) Next, in step S**102***c***3**, with reference to the port management table **171**, the switching program **121** of the switch unit **11** selects two ports having the lowest load among the plurality of ports **19** of the controller on the counterpart side (controller unit **15***a*).
(179) Next, in step S**102***c***4**, the switching program **121** of the switch unit **11** executes the switching processing shown in FIG. **8** on the two ports **19** selected in step S**102***c***3**. When the step S**102***c***4** ends, the switching program **121** of the switch unit **11** proceeds the processing to step S**102***d* (FIG.

14).

(180) Activation Processing of New Storage Controller According to Embodiment 2

(181) FIG. **16** is a sequence diagram showing activation processing of a new storage controller according to Embodiment 2.

(182) As compared with the activation processing of the new storage controller according to Embodiment 1 shown in FIG. **9**, the activation processing of the new storage controller according to Embodiment 2 is different in that steps S**140***a* to S**140***g* are executed between step S**140** and step S**141**. Except for the difference, the stop processing of the new storage controller according to Embodiment 2 is the same as the stop processing of the new storage controller according to Embodiment 1.

(183) In step S**140***a*, the control program **161** of the controller unit **15***a* starts switching processing of the connection route. Next, in step S**140***b*, the control program **161** of the controller unit **15***a* transmits the instruction to switch the connection route to the switching program **121** of the switch unit **11**.

(184) Next, in step S**140***c*, the switching program **121** of the switch unit **11** executes the low-load determination and connection route selection processing. The low-load determination and connection route selection processing is the same as that in step S**102***c* (FIG. **14**) and is as shown in FIG. **15**.

(185) In step S**140***c*, instead of selecting two ports **19** having the lowest load, two ports **19** related to the connection route with the "presence of conversion" for the "presence or absence of conversion" in the route management table **131** may be selected. In step S**140***c*, the two ports **19** related to the connection route with the "presence of conversion" for the "presence or absence of conversion" in the route management table **131** are selected, and the subsequent processing is executed, so that the connection route updated in step S**102***d* can be returned to the connection route before the update. For example, as shown in FIGS. **13**A and **13**C, the connection route of the input port **14**A (port #**4**) to the output port **14**B (port #**11**) is set by the port conversion of WWN #**4**.fwdarw.WWN #**3**. The connection route can be restored to the connection route of the input port **14**A (port #**4**) to the output port **14**B (port #**12**) based on the conversion information of WWN #**4**.fwdarw.WWN #**3**.

(186) Next, in step S**140***d*, the switching program **121** of the switch unit **11** reflects the connection route selected in step S**140***c* in the route management table **131** to update the route management table **131**. Next, in step S**140***e*, the switching program **121** of the switch unit **11** transmits a notification of update completion of the connection route to the control program **161** of the controller unit **15***a*. The notification of update completion of the connection route includes port information selected in step S**140***c*.

(187) Next, in step S**140***f*, the control program **161** of the controller unit **15***a* switches the host path definition based on the notification of update completion of the connection route notified in step S**140***e*, and updates the port management table **171**.

Effects of Embodiment 2

(188) In Embodiment 2 described above, two ports, that is, the port having the lowest load and the port having the second lowest load, are selected from the ports **19** in the controller unit **15***a*. The IO input to the first port having the lowest load and the IO input to the second port having the second lowest load among the two ports are aggregated to be input to the second port. That is, a free port is created in one controller unit **15**. Second address information is converted to the address information related to the first port as first address information. The first host path definition is set for the free port in the port management table **171** based on the address information related to the first port, and the first connection route is set in the route management table **131** based on the address information related to the first port.

(189) Thus, according to Embodiment 2, a transfer load can be equalized by adjusting the load in the controller unit **15** based on a confirmation result of the IO load of the port to create the free port

and selecting the free port of the switch unit **11** to switch the connection route. Since the transfer load is equalized, a transfer and stop time of the controller unit **15** accompanying the upgrading of the controller unit **15** can be shortened, and the work can be quickly performed.

(190) Although the embodiments according to the present disclosure have been described in detail above, the present disclosure is not limited to the embodiments described above, and various modifications can be made without departing from the gist of the present disclosure. For example, the embodiments described above are described in detail for easy understanding of the invention, and the invention is not necessarily limited to those including all the configurations described above. A part of the configuration of each embodiment may be added, deleted, or replaced with other configurations.

(191) Each of the configurations, function units, processing units, processing methods or the like described above may be implemented by hardware by designing a part or all of them with, for example, an integrated circuit. The above configurations, functions, or the like may also be implemented by software by interpreting and executing a program for implementing respective functions by the processor. Information such as a program, a table, and a file for implementing each function may be placed in a recording device such as a memory, an HDD, and an SSD, or a recording medium such as an IC card, an SD card, and a DVD.

(192) The drawings described above show control lines and information lines as considered necessary for description, but do not show all control lines and information lines in the products. For example, it may be considered that almost all the configurations are actually connected to each other.

(193) The function of each system and each device described above and the arrangement form of data are merely examples. The arrangement form of each function and data may be changed to an optimal arrangement form from a viewpoint of the performance of hardware or software, a processing efficiency, a communication efficiency, and the like.

Claims

1. A storage controller comprising: a switch unit configured to receive an IO from a host; a first storage control device and a second storage control device configured to process the IO received by the switch unit and to output and receive data to and from a disc device, wherein the switch unit includes an input port configured to receive the IO from the host, and an output port configured to output the IO to a port of the first storage control device and a port of the second storage control device; a storage unit that stores port management information for managing host path definition information for each of the first storage control device and the second storage control device, the host path definition information indicating a correspondence relation between the host and address information held by the port of a transmission destination to which the host transmits the IO, and route management information for managing a connection route for connecting the input port to the output port connected to the port holding the address information in the switch unit to transfer the IO; and a processor configured to receive an instruction to stop the second storage control device, select, from a plurality of ports of the first storage control device, two ports including a first port having a lowest load and a second port having a second lowest load, aggregate a first IO input from the first port and a second IO input from the second port such that both the first IO input and the second IO input are input to the second port, convert second address information to first address information, the second address information indicating the address information held by the port of the second storage control device, and the first address information indicating the address information related to the first port of the first storage control device, set, in the port management information, a first host path definition indicating the host path definition information between the host and the first address information in the first storage control device, in addition to a second host path definition indicating the host path definition information between the host and the second

address information in the second storage control device, wherein the first host path definition in the port management information is set based on the address information related to the first port, set, in the route management information, a first connection route indicating the connection route between the input port and a first output port serving as the output port to which the port holding the first address information is connected, in addition to a second connection route indicating the connection route between the input port and a second output port serving as the output port to which the port holding the second address information is connected, wherein the first connection route in the route management information is based on the address information related to the first port, and control the switch unit based on the route management information and control the first storage control device and the second storage control device based on the port management information to transfer the IO to the first storage control device or the second storage control device.

2. A storage controller comprising: a switch unit configured to receive an IO from a host; a first storage control device and a second storage control device configured to process the IO received by the switch unit and to output and receive data to and from a disc device, wherein the switch unit includes: an input port configured to receive the IO from the host, and an output port configured to output the IO to a port of the first storage control device and a port of the second storage control device; a storage unit that stores: port management information for managing host path definition information for each of the first storage control device and the second storage control device, the host path definition information indicating a correspondence relation between the host and address information held by the port of a transmission destination to which the host transmits the IO, and route management information for managing a connection route for connecting the input port to the output port connected to the port holding the address information in the switch unit to transfer the IO; and a processor configured to: receive an instruction to stop the second storage control device, convert second address information to first address information, the second address information indicating the address information held by the port of the second storage control device, and the first address information indicating the address information held by the port of the first storage control device, set, in the port management information, a first host path definition indicating the host path definition information between the host and the first address information in the first storage control device, in addition to a second host path definition indicating the host path definition information between the host and the second address information in the second storage control device, set, in the route management information, a first connection route indicating the connection route between the input port and a first output port serving as the output port to which the port holding the first address information is connected, in addition to a second connection route indicating the connection route between the input port and a second output port serving as the output port to which the port holding the second address information is connected, record, in a management table, identification information of the IO that is input to the input port before the first host path definition and the first connection route are set and that is not completely processed by the second storage control device, determine whether the identification information of the IO is recorded in the management table for each IO, output the IO to the second storage control device via the second connection route when the identification information of the IO is recorded in the management table, output the IO to the first storage control device via the first connection route when the identification information of the IO is not recorded in the management table, and control the switch unit based on the route management information and control the first storage control device and the second storage control device based on the port management information to transfer the IO to the first storage control device or the second storage control device.

3. The storage controller according to claim 2, wherein the processor is further configured to: when the IO whose identification information is recorded in the management table no longer exists delete the second host path definition from the port management information, and delete the second connection route from the route management information.

4. The storage controller according to claim 3, wherein the processor is further configured to: receive an instruction to activate the second storage control device, restore, in the port management information, the second address information from the first host path definition based on conversion information between the second address information and the first address information, restore, in the route management information, the second connection route from the first connection route based on the conversion information between the second address information and the first address information, and transfer the IO to the first storage control device or the second storage control device according to the port management information and the route management information.
5. The storage controller according to claim 4, wherein the processor is further configured to: record, in the management table, identification information of the IO that is input to the input port before the second host path definition and the second connection route are restored and that is not completely processed by the first storage control device, determine whether the identification information of the IO is recorded in the management table for each IO, output the IO to the first storage control device via the first connection route when the identification information of the IO is recorded in the management table, and output the IO to the second storage control device via the second connection route when the identification information of the IO is not recorded in the management table.
6. The storage controller according to claim 5, wherein the processor is further configured to: when the IO whose identification information is recorded in the management table no longer exists delete the first host path definition from the port management information, and delete the first connection route from the route management information.
7. A storage controller control method to be executed by a storage controller, wherein the storage controller includes: a switch unit configured to receive an IO from a host, a first storage control device and a second storage control device configured to process the IO received by the switch unit and to output and receive data to and from a disc device, a storage unit, and a processor, wherein the switch unit includes an input port configured to receive the IO from the host, and an output port configured to output the IO to a port of the first storage control device and a port of the second storage control device, wherein the storage unit stores port management information for managing host path definition information for each of the first storage control device and the second storage control device, the host path definition information indicating a correspondence relation between the host and address information held by the port of a transmission destination to which the host transmits the IO, and route management information for managing a connection route for connecting the input port to the output port connected to the port holding the address information in the switch unit to transfer the IO, and the storage controller control method comprises: receiving an instruction to stop the second storage control device; selecting, from a plurality of ports of the first storage control device, two ports including a first port having a lowest load and a second port having a second lowest load; aggregating a first IO input from the first port and a second IO input from the second port such that both the first IO input and the second IO input are input to the second port; converting second address information to first address information, the second address information indicating the address information held by the port of the second storage control device, and the first address information indicating the address information related to the first port of the first storage control device; setting, in the port management information, a first host path definition indicating the host path definition information between the host and the first address information in the first storage control device, in addition to a second host path definition indicating the host path definition information between the host and the second address information in the second storage control device, wherein the first host path definition in the port management information is set based on the address information related to the first port; setting, in the route management information, a first connection route indicating the connection route between the input port and a first output port serving as the output port to which the port holding the first address information is connected, in addition to a second connection route indicating the connection route

between the input port and a second output port serving as the output port to which the port holding the second address information is connected, wherein the first connection route in the route management information is based on the address information related to the first port; and controlling the switch unit based on the route management information and controlling the first storage control device and the second storage control device based on the port management information to transfer the IO to the first storage control device or the second storage control device.